Ryan Fernandes

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Experience

Associate Software Engineer July 2025 - Present
Aatmunn Software Development Center

- Owned end-to-end development and production deployment of an AI conversational agent serving 90,000+ global users on the Aatmunn platform, enabling natural language navigation across 55 screens and automated execution of tasks including user management, assignments, and data summarization.
- Architected a microservices system comprising a FastAPI agent service and FastMCP tool server, integrated with a Java backend via bearer token authentication and Postgres-based conversation state management.
- Built a RAG-based navigation engine using pgvector and a dual-table vector embedding system for semantic screen matching across 55 screens, routing user queries to relevant URLs with high precision.
- Implemented a ReAct agent (LangChain/LangGraph) with a supervisor node that generates a step-by-step execution plan per query, enabling simultaneous routing to both navigation and task execution, supporting complex multi-intent requests in a single interaction, compared to the prior single-path architecture.

Software Developer - Intern March 2025 - May 2025
CyberSafeHaven Technologies

- Developed the "Add Evidence" feature for Cyberisk4Board, allowing users to attach files and links to mandatory compliance questions, streamlining documentation for audits.
- Improved phishing domain detection in ThreatProwler by integrating GrayHat Warfare and DNSTwist, enabling scanning of suspicious or spoofed domain names.
- Refactored the partner data storage structure in ThreatProwler: replaced inefficient list-based JSON storage with key-based storage in Redis, enabling instant access to individual partner records by ID and eliminating the need for linear searches.

Projects

Retrieval-Augmented Generation (RAG) Chatbot with Real-Time Data March 2025

- Developed a RAG chatbot using Groq's LLM and FastAPI, integrating Chroma DB for semantic search to generate context-aware responses paired with relevant images.
- Implemented real-time web data retrieval using DuckDuckGo API and Beautiful Soup, enabling dynamic sourcing of up-to-date information for user queries.
- Designed and deployed a full-stack system with a Next.js frontend and Python backend.

Vision Transformer (ViT) Built from Scratch April 2025

- Implemented a full Vision Transformer architecture from first principles using PyTorch, including custom-built layers (e.g., Linear, LayerNorm, Attention).
- Developed key components: patch embedding, positional encoding, multi-head self-attention, and transformer encoder blocks without relying on external ViT libraries.
- Built a complete forward pass pipeline for image classification with handcrafted model components and minimal dependencies.

Technical Skills

Languages: Python, C/C++, MATLAB, PostgreSQL, HTML, CSS, JS
Frameworks: PyTorch, FastAPI, FastMCP, LangChain, LangGraph, LangSmith
Developer Tools: Git, GitHub, VS Code, Google Colab, Jupyter notebook, Docker
Libraries: Pandas, NumPy, BeautifulSoup, Scikit-learn

Education

KLE Technological University Vidyanagar, Hubballi
B.E in Electronics and Communications *2021 - 2025*

Prism PU College Dharwad, Karnataka
2019 – 2021